IIIT BHOPAL

Humor Detection in Hindi-English Code-Mixed Social Media Text Using Transformer Based Models

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Introduction & Dataset Overview

Background & Motivation

Social media is a primary channel of communication, where humor, irony and sarcasm keep conversations engaging. Detecting humor supports conversational AI, sentiment analysis and social network monitoring.

The Code-Mixing Challenge (Hinglish)

Users mix Hindi and English in one sentence using Roman script, slang, inconsistent spellings and shifting grammar, which makes traditional monolingual NLP models fail.

Project Aim

Develop and evaluate an XLM-RoBERTa framework that classifies social media text as Humorous or Non-Humorous.

DATASET: hindi_humour.csv

15,576

total rows (raw)

after removing missing values and empty entries

11,675

valid Hinglish samples for training and evaluation



Humorous

Literature Review / Background

Evolution of approaches and the code-mixed gap.

1

TRADITIONAL NLP

Handicraft rules ,Bag-of-Words
TF-IDF + Naive-Bayes

Mihalcea & Strapparava, 2005
Salton & Buckley, 1988

2

DEEP LEARNING

CNN's and LSTM's for
text classification

Kim, 2014
Chen and Lee, 2017

3

Transformers

Self-attentionandmodels
like BERT, XLM-R

Vaswani et al., 2017
Devlin et al., 2019
Conneau et al., 2020

4

CODE-MIXED GAP

Needfor Hinglishhumor
detection evaluation

Khandelwal et al., 2018
Khanuja et al., 2020

Research has advanced from rule-based methods to transformer models, yet **Hinglish humor detection** remains an under-explored niche.

Problem Statement & Objectives

Problem Definition

Building an automated text classification model to determine whether a given Hinglish (Hindi-English code-mixed) social media sentence is Humorous or Non-Humorous.

Key Challenges

- **Mixed Languages & Scripts:** Users mix Hindi and English words together using Roman script, making standard dictionary lookups fail.
- **Spelling Variations:** The same word can be spelled in multiple casual ways (e.g., "tumhaaree" vs. "tumhari").
- **Conflicting Data Labels:** About 19.5% of the dataset rows contain label conflicts, where identical sentences are sometimes marked as funny and other times as serious.
- **Missing Dialogue Context:** Sentences are analyzed entirely on their own, without knowing what was said right before them in a conversation.

Project Objectives

1. **Data Preparation & Cleaning :**
Load and clean the raw hindi_humour.csv dataset, eliminate missing values, and structure balanced data partitions.

2. **Transformer Fine-Tuning:**
Adapt the pretrained multilingual transformer (XLM-RoBERTa-base) by adding a custom classification head to learn deep contextual patterns.

3. **Rigorous Evaluation & Benchmarking :**
Evaluate model performance using robust metrics (Accuracy, Precision, Recall, and Weighted F1-score) and establish a comparative benchmark against classical ML baselines (Naive Bayes, Logistic Regression, Linear SVM).

Proposed Methodology

1 Dataset Loading

Raw CSV input: 15,576 rows

2 Preprocessing

Remove missing values, strip whitespace: 11,675 cleaned rows

3 Stratified Split

80% train (9,340), 10% validation (1,167), 10% test (1,168)

4 Tokenization

SentencePiece subwords (xlm-roberta-base), max length 128

5 Model Training

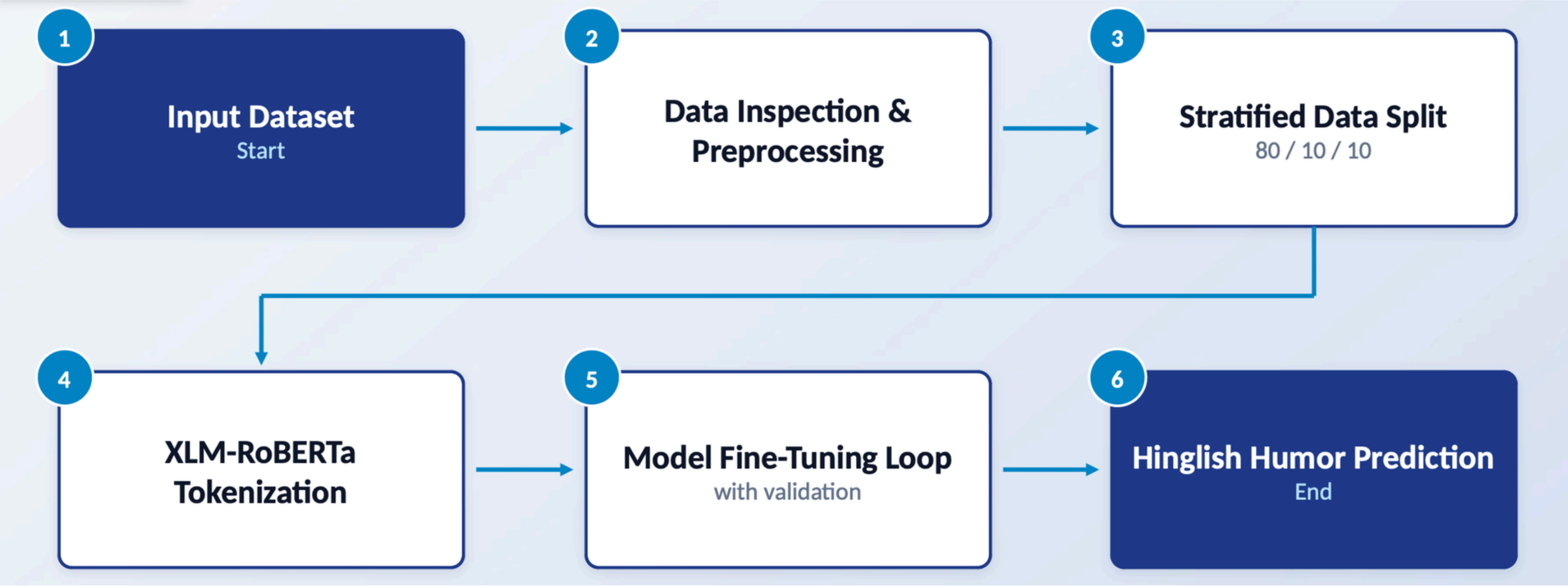
Fine-tune XLM-R with a 2-class classification head

6 Evaluation & Prediction

Accuracy and weighted F1-score on the test set

Why XLM-RoBERTa? Pre-trained on 100 languages, it provides a shared embedding space for processing mixed Hindi-English vocabulary.

System Diagram / Workflow



Implementation Details

Experimental Environment

PLATFORM

Google Colab (Cloud Notebook)

HARDWARE

NVIDIA Tesla T4 GPU

FRAMEWORKS

Python 3, PyTorch, Hugging Face Transformers

Training Configuration

Parameter	Value
Pretrained Model	xlm-roberta-base
Learning Rate	2×10^{-5}
Batch Size	16 (train & eval)
Epochs	4
Optimizer	AdamW, weight decay 0.01
Precision	Mixed precision (fp16)

Results, Comparison & Analysis

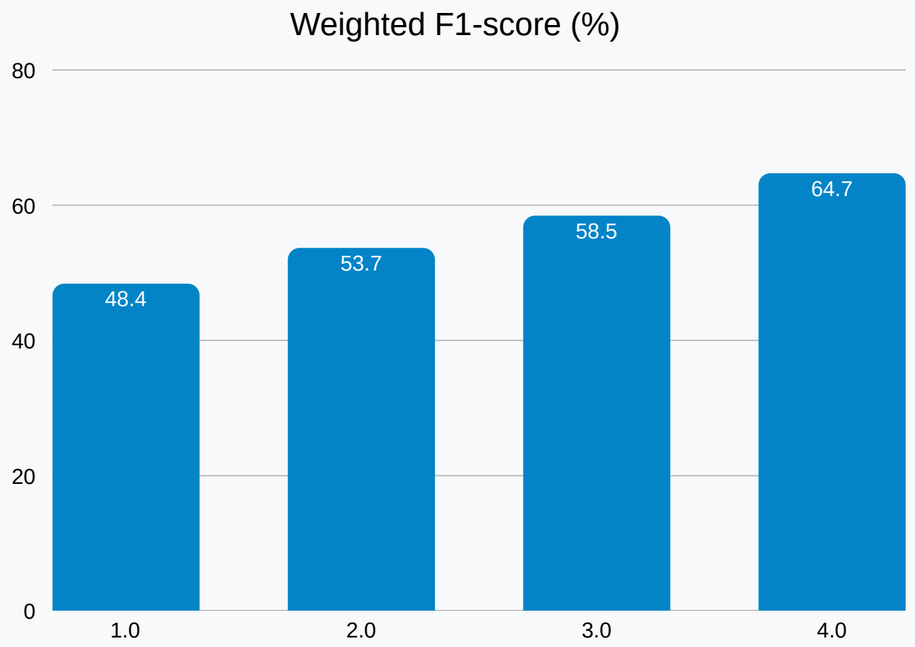
64%

Test Accuracy

64.72%

Weighted F1-Score

Model	Accuracy	Weighted F1
Majority-Class Guess	62.75%	48.39%
Naive Bayes + TF-IDF	63.26%	53.68%
Logistic Regression + TF-IDF	62.11%	58.45%
XLM-RoBERTa (Proposed)	**64.00%**	**64.72%**



Key Takeaway: Accuracy gains over the majority baseline are small, but the much higher weighted F1 shows the transformer detects humorous content instead of just predicting the majority class.

Conclusion & Future Scope

Conclusion

Successful Model Adaptation:

- Fine-tuned XLM-RoBERTa-base for binary humor classification on Hindi-English code-mixed (Hinglish) social media text, achieving 64% accuracy and a 64.72% weighted F1-score on unseen test data.

Superiority Over Classical Baselines:

- Significantly outperformed classical TF-IDF models (Naive Bayes 53.68%, Logistic Regression 58.45%, Linear SVM 58.39%) in weighted F1-score, proving that transformer self-attention detects humor far more evenly across both classes rather than collapsing to the majority class.

Identification of Primary Bottlenecks:

- Discovered that overall accuracy is heavily capped by dataset noise (~19.5% conflicting labels) and the absence of prior conversational context, rather than model architectural limits.

Future Scope

Data Reannotation & Curation:

- Manually resolve label conflicts, remove cross-split duplicates, and implement script normalization (standardizing Devanagari vs. Romanized spellings like "tumhari" and "tumhaaree").

Multi-Turn Context Integration:

- Feed preceding dialogue turns alongside the target sentence to capture conversational setup and context-dependent humor.

Model Benchmark Expansion:

- Evaluate and compare performance against Indian language pre-trained models such as MuRIL, IndicBERT,mBERT, and XLM-RoBERTa-large.

Practical System Deployment:

- Wrap the trained transformer pipeline into a REST API / web application for real-time automated social media content moderation and conversational AI integration.

References

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HUMOR DETECTION IN HINDI-ENGLISH CODE-MIXED TEXT

Thank You

Questions are welcome.

CODE & REPOSITORY

</> github.com/Sanyam-Jain27/minor-project

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